MIL-M-19981A(SHIPS)
26 February 1962
SUPERSEDING
MIL-M-19981(SHIPS)
2 August 1957

MILITARY SPECIFICATION

METERS, AIR FLOW INDICATING, DIFFERENTIAL PRESSURE, SUBMARINE BATTERY VENTILATION, VARIABLE TRANSFORMER

1. SCOPE

1.1 Scope. - This specification covers submarine battery ventilation air flow indicating meters. The meters measure the differential pressure created by air flow through a venturi meter and transmit a proportional voltage to a remote indicator, which can be inscribed to indicate in cubic feet per minute (c.f.m.).

1.2 Classification. - Air flow indicating meters shall be of the following types as specified (see 6.1):

Type A - One transmitter and one remote indicator.
Type B - One transmitter and two remote indicators.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard. Application, Tests for.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-D-963 - Drawings, Electrical, Hull and Mechanical Equipment for Naval Shipboard Use.
MIL-Q-9858 - Quality Control System Requirements.
MIL-P-15024 - Plates, Identification —Information and Marking for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-M-15071 - Manual, Service (Instruction Book) for Shipboard Electrical and Mechanical Equipment.
MIL-M-16034 - Meters, Electrical—Indicating (Switchboard Portable Types)
MIL-R-18281 - Rectifiers, Metallic: Selenium, Copper Oxide, and Magnesium-Copper Sulfide Types, Naval Shipboard Use.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd Street, New York 16, New York.)

3. REQUIREMENTS

3.1 Preproduction model. - Prior to production, a preproduction model shall be submitted for the tests specified in 4.2.

3.2 Material. - Material shall be in accordance with MIL-E-917, as applicable to the design, material and construction.

3.3 Major components. - Air flow indicating meters shall consist of two major components; a

FSC 6680

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transmitter assembly and one or two remote indicator assemblies, as applicable (see 3.5). The transmitter assembly shall consist of a variable voltage output transformer operated by a diaphragm. The diaphragm will be connected across the high and low static pressure taps of a venturi meter located in the battery ventilation exhaust duct. The pressure differential of the venturi meter, at maximum air flow, will be approximately 4 inches water, gage. The remote indicator shall be a voltmeter which indicates the output of the variable voltage transformer in c.f.m. The change of the remote indicator deflection shall be approximately linear to the change of the pressure differential.

3.4 Transmitter assembly. -

3.4.1 An adjustment shall be provided whereby the maximum output of the variable transformer can be varied to give maximum deflection on the remote indicator for any variation of venturi meter pressure from 3-1/2 to 4-1/2 inches of water, gage.

3.4.2 A common terminal board shall be provided for all electrical connections. The terminal board shall be protected within the transmitter assembly or in a connection box and shall be located such that all connections can be made outside the airtight or otherwise sealed portions of the transmitter assembly.

3.4.3 The transmitter assembly shall be designed for mounting on a bulkhead or the overhead.

3.5 Indicator. -

3.5.1 Type A. - Type A. - The type A remote indicator shall consist of a direct current (d.c.) voltmeter conforming to MIL-M-16034 with a blank scale. There shall be a rectifier conforming to MIL-R-18281, to convert the alternating current (a.c.) output of the transmitter to suit the d.c. voltmeter. The entire remote indicator assembly shall be designed to fit a panel space 4-1/2 by 4-1/2 inches by 3-7/8 inches deep.

3.5.2 Type B. - The type B remote indicator shall consist of two d.c. voltmeters, conforming to MIL-M-16034, connected in parallel. The requirements of 3.5.1 shall be applicable to each remote indicator.

3.6 Shock. - Airflow indicating meters shall be capable of withstanding the HI shock test specified in MIL-S-901.

3.7 Vibration. - Airflow indicating meters shall be capable of withstanding the vibration tests specified by type I of MIL-STD-167.

3.8 Accuracy. -

3.8.1 The error of flow indication shall not exceed 5 percent of full scale when the airflow indicating meter is rolled 30 degrees in any direction.

3.8.2 The error of flow indication shall not exceed 2 percent of full scale when the airflow indicating meter is subjected to temperature fluctuations from 60° to 130° F.

3.8.3 The error of flow indication shall not exceed 5 percent of full scale for a power supply variation of plus or minus 7 volts or plus or minus 5 cycles per second.

3.8.4 The error of flow indication shall not exceed 5 percent of full scale when the airflow indicating meter is subjected to a linear acceleration of 8 feet per second per second in the directions normal to, and parallel to the diaphragm.

3.8.5 The error of flow indication shall not exceed 5 percent of full scale when the airflow indicating meter is subjected to an angular acceleration created by a roll about any axis to 60 degrees from normal and back to the normal position in a period of 12 seconds.

3.9 Pressure connections to the transmitter assembly shall be 1/2 iron pipe size pipe connections.

3.10 Electrical insulation. - Electrical insulation shall be class A of MIL-E-917, or better.

3.11 Painting. - All external parts shall be painted in accordance with MIL-E-917. All internal parts shall be of corrosion-resistant material or otherwise suitably provided against corrosion.

3.12 Identification plates. - Identification plates shall be provided in accordance with MIL-P-15024. Identification plates shall include the following:

(a) Name of instrument and component.
(b) Serial number.
(c) Stock number.
(d) Contract number.
(e) Name and address of manufacturer.
(f) Electrical data (voltage, cycles, and phase).
(g) Ampere power consumption.

A small identification plate shall be provided on the remote indicator to identify it with the basic instrument.

3.13 Drawings. - Class A or C and Class B drawings shall be provided in accordance with

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MIL-D-963 and as specified herein. The drawings shall include certification data and assembly drawings. Approval shall be by the bureau or agency concerned.

3.13.1 The assembly drawing shall show outline, mounting, attachment and connection dimensions including methods and sizes of fastenings and clearances for installation and servicing plus supplementary data as necessary to permit shipyard installation without suppliers assistance. The drawing shall illustrate design, construction, operation, and identity of parts. Performance data shall be included. Acceptance tests shall be identified and approval authority and data should be noted on the drawing. The drawing shall be sufficiently complete to reflect compliance with the requirements of this specification for the equipment. Subassembly drawings may be provided to supplement assembly drawings, where desirable.

3.13.2 Certification data shall be in accordance with MIL-D-963 and shall include under "Additional Data" a list of specification acceptance and shock tests conducted on the equipment.

3.14 Manuals. - Manuals shall be in accordance with MIL-M-15071.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 The contractor shall provide and maintain a quality control system acceptable to the Government for the supplies covered by this specification. The system of quality control shall be in accordance with MIL-Q-9858.

4.2 Preproduction inspection. - Prior to production, the first unit under the contract or order shall be subjected to the tests specified in 4.6. The unit shall successfully pass these tests prior to production of the remaining units. Any modification found necessary as a result of the tests shall be applied to the remaining units.

4.2.1 Where a manufacturer has previously conducted shock and vibration tests in accordance with the requirements of this specification and the units have been accepted by the bureau or agency concerned, the tests need not be repeated and a test report of the initial tests will be acceptable.

4.3 Sampling. -

4.3.1 Lot. - All meters of the same type, offered for delivery at one time, shall be considered a lot.

4.3.2 Sampling for group A examination and tests. - A random sample of meters shall be selected from each lot in accordance with inspection level III of MIL-STD-105 for the group A examination and tests specified in 4.4.1. The acceptable quality level shall be 1.5 percent defective.

4.3.3 Sampling for group B tests. - A random sample of meters shall be selected from each lot in accordance with inspection level L-6 of MIL-STD-105 for the tests specified in 4.4.2. If any meter fails in any test, the lot represented by the sample shall be rejected.

4.4 Acceptance inspection. -

4.4.1 Group A examination and test. - Each of the meters selected in accordance with 4.3.2 shall be subjected to each of the following examination and test and the results compared with the requirements of this specification:

Group A examination and test	Reference paragraph
Examination	4.5
Accuracy at zero and 4 inches pressure	4.6.1

4.4.2 Sampling for group B tests. - Each of the meters selected in accordance with 4.3.3 shall be subjected to each of the following tests and the results compared with the requirements of this specification:

Group B tests	Reference paragraph
Accuracy at zero and 4 inches pressure	4.6.1
Effect of rotation	4.6.2 and 4.6.6
High and low temperature	4.6.3
Variation of supply line voltage and frequency	4.6.4
Effect of acceleration	4.6.5

4.5 Examination. - Each of the meters selected in accordance with 4.3.2 shall be examined to determine conformance with all the requirements of this specification not involving tests.

4.6 Tests. -

4.6.1 Accuracy at zero and 4 inches pressure. - With the airflow indicating meter level and connected

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to a 115 volt, 60 cycle, power supply, an air pressure differential of 4 inches water gage shall be applied across the transmitter and the indicator shall be adjusted to slightly less than full scale deflection. The pressure differential shall then be varied over the range zero to 4 inches and pressure differential versus indicator pointer deflection, in degrees, plotted.

4.6.2 Effect of rotation (30 degrees). - From the normal operating position, the transmitter assembly shall be rotated 30 degrees in both directions and about at least two horizontal axes 90 degrees apart in the horizontal plane. The battery airflow meters shall be operated at a minimum of two different pressures in each position, and the deviation from the curve plotted as specified in 4.6.1 shall not exceed the limit specified in 3.8.1.

4.6.3 High and low temperature. - The indicator deflection shall be observed for two different pressure differentials with the system at 60°F. and at 130°F. and at one intermediate temperature. Temperature equilibrium shall be reached for each test. The deviation of the indicator pointer from the calibration curve shall not exceed the requirements specified in 3.8.2.

4.6.4 Variation of supply line voltage and frequency. - While maintaining a constant impressed pressure differential, the voltage shall be varied from 108 to 122 volts and the frequency varied from 55 cycles to 65 cycles per second. In each case the remote indicator pointer deviation from the position at 115 volts shall not exceed the requirement in 3.8.3.

4.6.5 Effect of acceleration. - The airflow indicating meter shall be linearly accelerated at a constant rate of 8 feet per second per second in two planes, one normal to the diaphragm, and one parallel to the diaphragm. The error of flow indication shall not exceed that specified in 3.8.4.

4.6.6 Effect of rotation (60 degrees). - With the remote indicator in a continually visible position, the meter shall be rolled about the normal position to an axis 60 degrees from the normal and back in a period of 12 seconds. The test shall be repeated about an axis 90 degrees removed. The error of flow indication shall not exceed that specified in 3.8.5.

4.6.7 Shock test. - One battery airflow meter of each design and manufacture shall be shock tested to determine conformance with 3.6.

4.6.8 Vibration test. - One battery airflow meter of each design and manufacture shall be tested to determine conformance with 3.7.

5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation.

5.1.1 Preservation and packaging. - Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early installation.

5.1.1.1 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers and method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation.

5.1.1.2 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and destination. (Special marking requirements for radio-active materials, magnetic, low-noise equipment, and so forth, shall be added when required.)

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing and marking for shipment shall be as specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Preservation and packaging. -

5.2.1.1.1 Level A. - The meter assembly, including remote indicator(s), shall be individually preserved and packaged in accordance with method II-B of MIL-P-116. Contact preservative is not required.

5.2.1.1.2 Level C. - The meter assembly shall be preserved and packaged in accordance with the manufacturer's commercial practice.

5.2.2 Packing. -

5.2.2.1 Level A. - The meter assembly, preserved and packaged in accordance with level A or C

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Marking. - In addition to any special marking required in the contract or order or herein, each intermediate package and shipping container shall be marked in accordance with MIL-STD-129).

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dering data. - Procurement documents should specify the following:

Title, number and date of this specification.
Type required (see 1.2)
Preservation, packaging, packing and marking requirements, if other than those required in 5.1 (see 5.2).

- When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government hereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing activity:
Navy - Ships
(Project 6680-N034Sh)

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SPECIFICATION ANALYSIS SHEET
NAVSHIPS-4863 (8-61)

BUDGET BU. NO. 45-R309

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Bureau of Ships

This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost.

Comments and the return of this form will be appreciated.

Fold on dotted lines on reverse side, staple in corner, and send to Bureau of Ships, Specifications and Standardization Branch, Washington 25, D.C.

SPECIFICATION

ORGANIZATION	CITY	STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A DIRECT GOVERNMENT CONTRACT ☐ OR A SUBCONTRACT ☐

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   a. GIVE PARAGRAPH NUMBER AND WORDING

   b. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID.

3. IS THE SPECIFICATION RESTRICTIVE? ☐ YES ☐ NO IF THE ANSWER IS "YES", IN WHAT WAY?

4. REMARKS (*Attach any pertinent data which may be of use in improving this specification.*) PLACE THIS FORM AND PAPERS IN AN ENVELOPE AND SEND TO THE BUREAU.

SUBMITTED BY (*Print name and activity*)	DATE

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